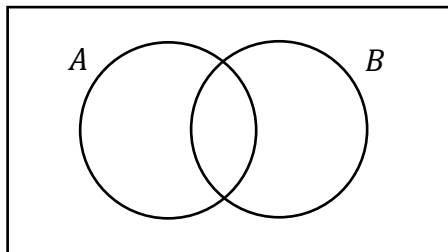


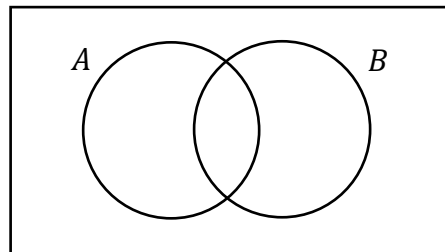
Discrete Mathematics: Foundations revision

- 1 The diagrams below illustrate a universal set $\mathcal{E}$ together with sets A and B . In each case shade the region indicated:

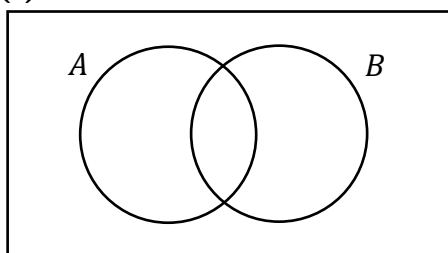
(a) $A \cap B'$



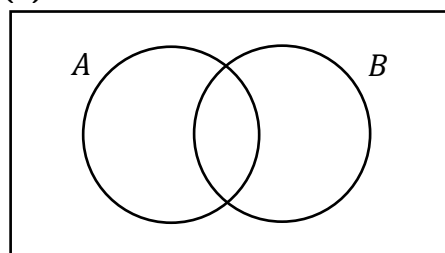
(b) $A' \cup B$



(c) $A' \cap B'$



(d) $A' \cup B'$



- 2 Given that $A \subset B$, simplify

(a) $A \cap B$

(b) $A \cup B$

(c) $A \cap B'$

(d) $A' \cup B$

- 3 Use the inclusion-exclusion principle to find the number of multiples of 2 or 3 in the set of integers from 1 to 100 inclusive.
- 4 The Head Master is feeling generous. He puts the 720 pupils in School into a list, and awards the book 'Twelfth Night' to every twelfth pupil. Additionally, he awards the book 'Catch-22' to every twenty-second pupil. How many pupils in total receive at least one book from the Head Master?
- 5 (a) How many different arrangements of the letters PATHOGEN are there?
- (b) How many different arrangements of the letters PATHOGEN are there in which the three vowels are together?
- 6 (a) How many different arrangements of the letters REPEATED are there?
- (b) How many different arrangements of the letters REPEATED are there in which the four consonants are together?

- 7 How many five-digit numbers are there in which the digits are in strictly increasing order?
- 8 How many six-digit even numbers can be made by rearranging the numbers 012345?
- 9 (a) How many different selections of five letters from MACHINE are there (the order in which the letters are selected doesn't matter)?
- (b) How many different selections of five letters from MACHINE are there which have exactly two consonants?
- 10 A committee of seven people needs to be chosen from a group of ten women and eight men.
- (a) How many different possible committees are there?
- (b) How many committees are there that consist of four women and three men?
- (c) How many committees are there that have at least one woman and at least one man?
- 11 (a) How many ways are there of splitting a group of nine people into two teams of sizes four and five?
- (b) How many ways are there of splitting a group of nine people into three teams each of size three?
- 12 How many different arrangements of the letters COFFEE are there in which there is no repeated letter? (That is, the two Es cannot be together, and the two Fs cannot be together.)
- 13 (a) How many different patterns can be made when two blue balls, two red balls, a yellow ball, and a green ball are put in a line?
- (b) How many different patterns can be made when two blue balls, two red balls, a yellow ball, and a green ball are put in a line if no balls of the same colour are placed next to each other?
- 14 A triangle has side length 2 cm, and five points are chosen inside the triangle. Use the pigeonhole principle to show that at least two of these points are no more than 1 cm apart.
- 15 Use the pigeonhole principle to show that for every positive integer n there exists a number divisible by n that consists only of the digits 0 and 5.
[Hint: Consider the sequence 5, 55, 555, 5555,]
- 16* Jon has seven cards with the numbers 0, 0, 2, 3, 4, 5, 7 on them. How many different five-digit even numbers can be made by using five of Jon's cards?